

Q-Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables

To observe how the RIP hop count metric changes when an additional router is inserted between two existing routers in a network topology.

Configure RIP version 2 on the routers.

Add an extra router between two existing routers.

Allow RIP to recalculate the routes.

Observe the hop count in the routing tables.

Compare the hop count before and after adding the extra router.

3. Network Topology

Before Adding the Extra Router

LAN 1 — R1 ——— R2 ——— R3 — LAN 3

The path from R1 to a network behind R3 was:

R1 → R2 → R3

After Adding the Extra Router

An additional router, R4, was inserted between R1 and R2:

LAN 1 — R1 ——— R4 ——— R2 ——— R3 — LAN 3

The new path became:

R1 → R4 → R2 → R3

Configuration

RIP version 2 was configured on the routers using the following commands:

```
enable
configure terminal
router rip
version 2
no auto-summary
```

The directly connected networks were advertised using the appropriate network commands.

The interfaces connecting the routers were assigned IP addresses and enabled using:

```
ip address <IP-address> <subnet-mask>
no shutdown
```

Verification

After allowing time for RIP convergence, the routing tables were checked using:

```
show ip route
```

RIP-learned routes were specifically checked using:

```
show ip route rip
```

The RIP configuration was verified using:

```
show ip protocols
```

Observation

The routing table displays RIP routes in the following general format:

```
R  <network> [120/<hop-count>] via <next-hop>
```

The value after 120/ represents the RIP hop count.

example:

```
R  192.168.30.0/24 [120/2] via ...
```

indicates a route with a metric of 2 hops.

After inserting R4, the route required one additional router to reach the destination.

Comparison

Destination Network	Before Adding R4	After Adding R4	Change
Network behind R2	1 hop	2 hops	+1
Network behind R3	2 hops	3 hops	+1

7. Result

After adding the extra router R4 between R1 and R2, the RIP hop count increased by **1** for routes that passed through the new router.

example:

Before:

R1 → R2 → R3

Hop count = 2

After:

R1 → R4 → R2 → R3

Hop count = 3

Therefore:

Increase in hop count = 3 – 2 = 1 hop

The experiment successfully demonstrated that RIP uses **hop count as its routing metric**. Each router crossed by a packet adds one hop to the RIP metric. When an additional router was inserted into the path, the routing tables were updated after RIP convergence and the hop count increased by one.

This confirms that adding routers to a path can increase the RIP metric and influence the route selected by RIP. RIP supports a maximum metric of **15 hops**, while a metric of **16** represents an unreachable destination.